CHAPTER 7

WEB HOSTING

CHAPTER ESSENCE AT A GLANCE:
Web hosting deploys developed digital web assets to globally interconnected 24/7 internet servers through ICANN domain registration, server storage provisioning, DNS record mapping, and secure FTP publishing.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Web Hosting Infrastructure and Lifecycle:** Web hosting reserves physical or virtual server storage, processing power, and high-bandwidth connectivity to make web pages continuously accessible worldwide via standardized domain addresses.

• **Domain Registration and DNS Zone Mapping:** Unique domain names purchased through accredited registrars link human-readable brand names to numerical IP addresses via DNS A-records, CNAME aliases, and MX mail exchange pointers.

• **Server Hosting Architectures:** Hosting architectures balance cost against dedicated performance, ranging from shared hosting with shared hardware, to VPS virtualization, dedicated bare-metal servers, and elastic scalable cloud infrastructure.

• **File Transfer Protocols and Modern Publishing:** Web assets are published to server directories like public html using File Transfer Protocol (FTP) clients such as FileZilla, designed responsively using viewport settings and Content Management Systems.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\text{Web Deployment Lifecycle} = \text{Domain Registration} \longrightarrow \text{Hosting Space} \longrightarrow \text{DNS Configuration} \longrightarrow \text{FTP Upload}$$

Site Publishing Deployment Stages

$$\text{Hosting Architecture Comparison} = \frac{\text{Shared (Low Cost, Shared CPU)} < \text{VPS (Isolated Virtual Core)}}{\text{Dedicated (Single Tenant, Full Bare Metal)} < \text{Cloud (Elastic Cluster, On-Demand)}}$$

Server Infrastructure Stratification

$$\text{DNS Zone Record Types} = \text{A Record} : \text{Name} \to \text{IPv4}, \quad \text{CNAME} : \text{Alias} \to \text{Canonical Name}, \quad \text{MX} : \text{Mail Routing}$$

Domain Zone File Mappings

$$\text{Responsive Viewport Rule} = <\text{meta name} = \text{"viewport" content} = \text{"width=device-width, initial-scale=1.0"} >$$

Mobile Responsive Viewport Directive

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Never confuse domain name registration with web hosting; domain registration reserves the name rights, while hosting rents physical server storage and bandwidth.

• **Alert:** Always upload website files into the designated public html or www root directory on the server, and verify that the main homepage file is strictly named index.html.

• **Alert:** In shared hosting, CPU, RAM, and bandwidth are shared among hundreds of websites; sudden traffic spikes on one site can severely degrade performance for all other tenants.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What is web hosting? Distinguish between domain name registration and web hosting. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Web Hosting: The commercial service of providing storage space and computing resources on specialized server computers connected $\frac{24}{7}$ to the Internet. **[1 Mark]**

• Distinction: Domain registration reserves the legal right to a unique alphanumeric web address (e.g. myschool.com), whereas web hosting leases the physical hard drive storage and server capacity where actual website files reside. **[1 Mark]**

2. What is the significance of the filename index.html on a web hosting server? [Score: 2 Marks]

Answer & Key Valuation Steps:

• Default Document: Web servers are configured by default to automatically deliver index.html as the primary landing homepage whenever a client accesses the root domain URL without specifying a file. **[1 Mark]**

• Directory Browsing Prevention: Having a valid index.html prevents unauthorized visitors from viewing raw directory listings of files hosted on the server. **[1 Mark]**

3. Compare Shared Hosting, Dedicated Hosting, and Virtual Private Server (VPS) hosting across cost, performance, and security. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Shared Hosting: Highly economical, multiple websites share the exact same physical server CPU, RAM, and disk storage; vulnerable to performance degradation caused by neighboring sites. **[1 Mark]**

• VPS Hosting: Uses software hypervisors to partition a physical machine into dedicated virtual servers with guaranteed CPU and RAM slices; balances cost and performance. **[1 Mark]**

• Dedicated Hosting: The entire physical server is leased exclusively to one organization; offers maximum computing power, custom software configuration, and isolated enterprise security at premium cost. **[1 Mark]**

4. Explain the role of FTP in web hosting. Describe how a client tool like FileZilla is used to transfer website files to a remote server. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Role of FTP: File Transfer Protocol is the network standard used to upload local HTML pages, images, and scripts securely to the web hosting server over port 21. **[1 Mark]**

• FileZilla Usage: The developer launches FileZilla, enters Host address, Username, Password, and Port into the Quickconnect bar to establish connection. **[1 Mark]**

• Transfer Process: Local project files in the left pane are selected and dragged across to the server public html folder in the right pane to complete deployment. **[1 Mark]**

5. What is Cloud Hosting? State its major advantages over traditional physical server hosting. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Concept: Websites are hosted across an interconnected cluster of virtualized cloud servers rather than on a single physical machine. **[1 Mark]**

• Elastic Scalability: Server resources (CPU, RAM, storage) dynamically scale up or down in real time to accommodate sudden traffic surges without downtime. **[1 Mark]**

• High Availability and Reliability: If any single hardware node experiences failure, another node in the cloud cluster immediately assumes workload seamlessly. **[1 Mark]**

6. What is Responsive Web Design (RWD)? Explain the role of the viewport meta tag and CSS media queries. [Score: 3 Marks]

Answer & Key Valuation Steps:

• RWD Concept: An approach where web pages dynamically adapt their layout, text sizing, and images to render optimally across smartphones, tablets, and desktop monitors. **[1 Mark]**

• Viewport Meta Tag: Configures the browser visible viewport width to match the physical device width (<meta name='viewport' content='width=device-width, initial-scale=1.0'>). **[1 Mark]**

• Media Queries: CSS rules (@media) that apply specific styling blocks conditionally based on screen resolution or orientation (e.g. max-width: 768px). **[1 Mark]**

7. Outline the complete step-by-step procedure for launching a school or business website on the internet, from selecting a domain name to live verification. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Step 1 (Domain Registration): Choose a unique memorable domain name and register it with an accredited registrar (e.g. GoDaddy, Namecheap) to reserve ownership. **[1 Mark]**

• Step 2 (Hosting Space Acquisition): Select an appropriate hosting package (Shared, VPS, Cloud) based on traffic estimates and purchase server space. **[1 Mark]**

• Step 3 (DNS Configuration): Point the domain Name Servers (NS) at the registrar portal to the hosting provider DNS servers to map the domain name to the host IP. **[1 Mark]**

• Step 4 (FTP Upload and Verification): Upload HTML, CSS, images, and index.html to public html using FileZilla, then test domain URL across multiple browsers to verify live availability. **[1 Mark]**